Crowd-Sourced Community Infrastructure Damage Mapping with Severity Prioritization

CSA0925-PROGRAMMING IN JAVA

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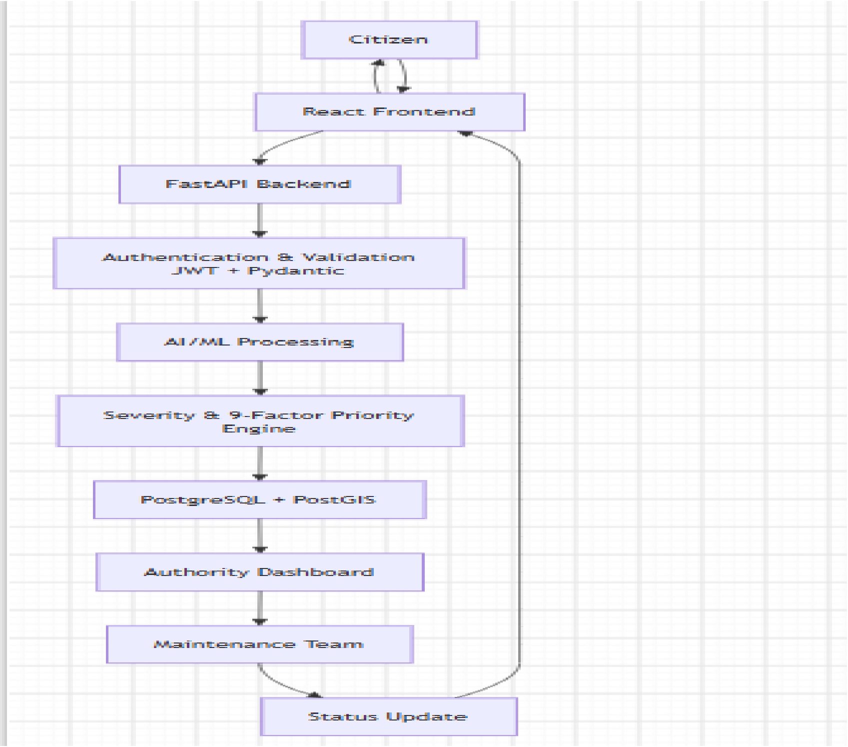
Review-1 Feedback

- **Feedback Received**
- Add detailed system architecture
- Clarify module responsibilities
- Specify Java-based technology stack
- **Corrective Actions**
- Updated architecture
- Tables created
- Defined module data flow
- Finalized Java technologies

Updated Architecture

➢ **Changes Made**

- Attractive Dashboard
- Login page for Citizen and Admin
- Added database layer
- Added severity scoring
- Added authority dashboard
- Added status notification flow



Module Design

1. Crowd-Sourced Reporting Module

Citizens report infrastructure issues.

Upload geotagged photos and descriptions.

Stores report and location details.

2. Severity Scoring & Heatmap Module

Analyzes damage severity and report density.

Uses AI/ML + 9-factor priority engine.

Generates priority heatmaps for authorities.

3. Authority Action Tracking Module

Authorities assign repair actions.

Maintenance teams update repair status.

Citizens receive status updates.

Data Flow:

Citizen → Crowd-Sourced Reporting→ Severity Scoring & Heatmap → Authority Action Tracking → Status Update → Citizen.

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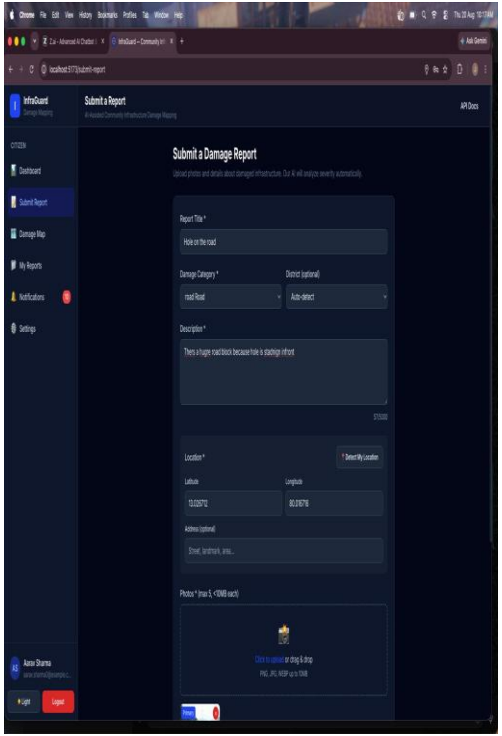
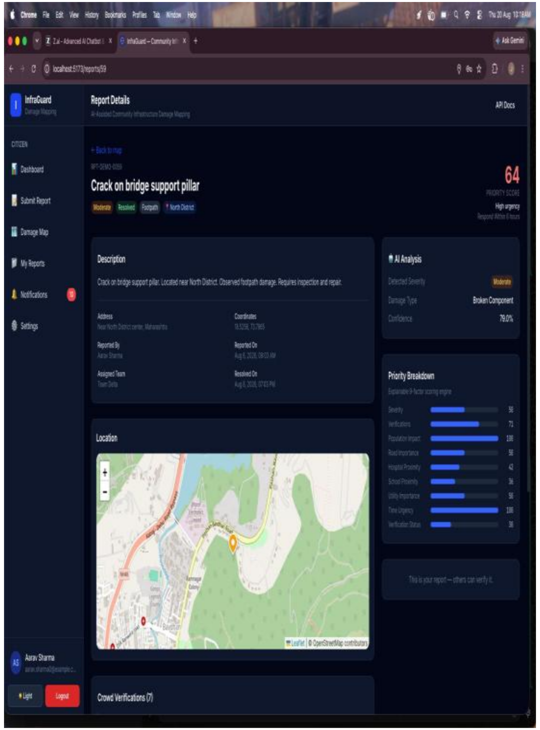
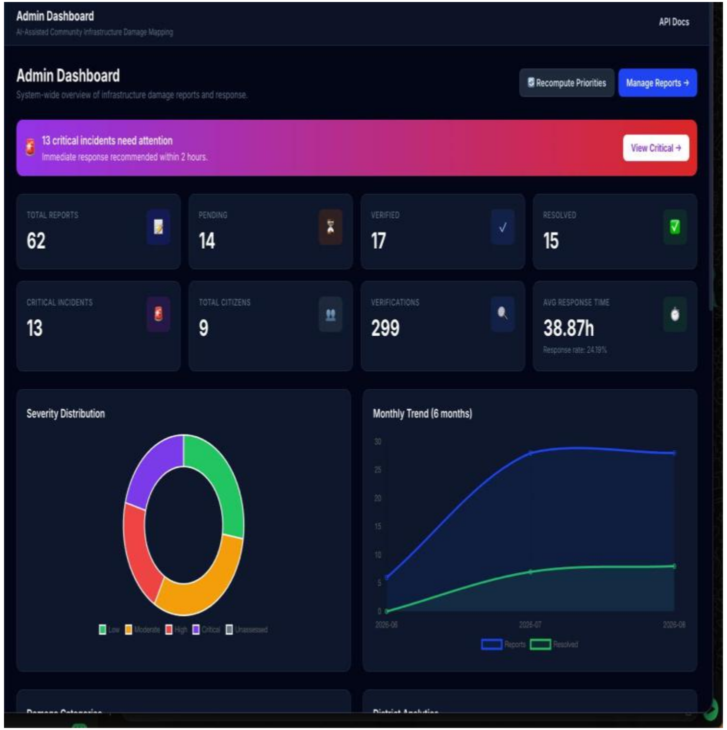
Modules Output







Technology Used

- **Frontend**
  - HTML
  - CSS
  - JavaScript
  - React
- **Backend**
  - Java 17
  - Spring Boot
  - Spring Web / REST API
  - Spring Data JPA
  - Hibernate
  - Maven

Authentication

JWT

Spring Security

AI / ML

If your AI model is implemented separately:

Python ML model

Java REST API integration

Tools

Eclipse / IntelliJ IDEA

MySQL Workbench

GitHub

Postman

draw.io

- **Database**
  - MySQL / PostgreSQL
  - JDBC
  - JPA / Hibernate

Implementation

Completion

- Overall: 70–80%

Implemented

- Java Spring Boot backend
- REST API development
- Database connectivity using JPA/Hibernate
- JWT-based authentication
- Crowd-sourced issue reporting
- Severity and priority calculation
- Authority action tracking
- Interactive map and heatmap

Code Summary

Users > lithish > college > CSA0925 > capstone > infraguard (1) > database > schema.sql

```sql
CREATE TABLE IF NOT EXISTS reports (
    id                      BIGSERIAL PRIMARY KEY,
    reference_code          VARCHAR(30) NOT NULL UNIQUE,
    user_id                 BIGINT NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    district_id             BIGINT REFERENCES districts(id) ON DELETE SET NULL,
    infrastructure_type_id  BIGINT NOT NULL REFERENCES infrastructure_types(id) ON DELETE RESTRICT,

    title                   VARCHAR(255) NOT NULL,
    description             TEXT NOT NULL,
    address                 VARCHAR(500),

    latitude                DOUBLE PRECISION NOT NULL,
    longitude               DOUBLE PRECISION NOT NULL,
    geom                    GEOMETRY(POINT, 4326),

    ai_severity             VARCHAR(20),
    ai_confidence           DOUBLE PRECISION,
    ai_damage_type          VARCHAR(100),
    ai_features             JSONB,

    final_severity          VARCHAR(20),

    status                  VARCHAR(30) NOT NULL DEFAULT 'Reported',
    credibility_score       DOUBLE PRECISION NOT NULL DEFAULT 0.0,

    verification_count      INTEGER NOT NULL DEFAULT 0,
    upvote_count            INTEGER NOT NULL DEFAULT 0,
    downvote_count          INTEGER NOT NULL DEFAULT 0,

    assigned_team           VARCHAR(150),
    resolution_notes        TEXT,
    resolved_at             TIMESTAMP,

    created_at              TIMESTAMP NOT NULL DEFAULT NOW(),
    updated_at              TIMESTAMP NOT NULL DEFAULT NOW(),

    CONSTRAINT ck_reports_status CHECK (
        status IN ('Reported','Verified','Rejected','Assigned','In Progress','Resolved')
    ),
    CONSTRAINT ck_reports_ai_severity CHECK (
        ai_severity IS NULL OR ai_severity IN ('Low','Moderate','High','Critical')
    ),
    CONSTRAINT ck_reports_final_severity CHECK (
        final_severity IS NULL OR final_severity IN ('Low','Moderate','High','Critical')
    )
);
CREATE INDEX IF NOT EXISTS idx_reports_status ON reports(status);
CREATE INDEX IF NOT EXISTS idx_reports_severity ON reports(ai_severity);
CREATE INDEX IF NOT EXISTS idx_reports_user ON reports(user_id);
CREATE INDEX IF NOT EXISTS idx_reports_district ON reports(district_id);
CREATE INDEX IF NOT EXISTS idx_reports_infra ON reports(infrastructure_type_id);
CREATE INDEX IF NOT EXISTS idx_reports_created ON reports(created_at DESC);
```

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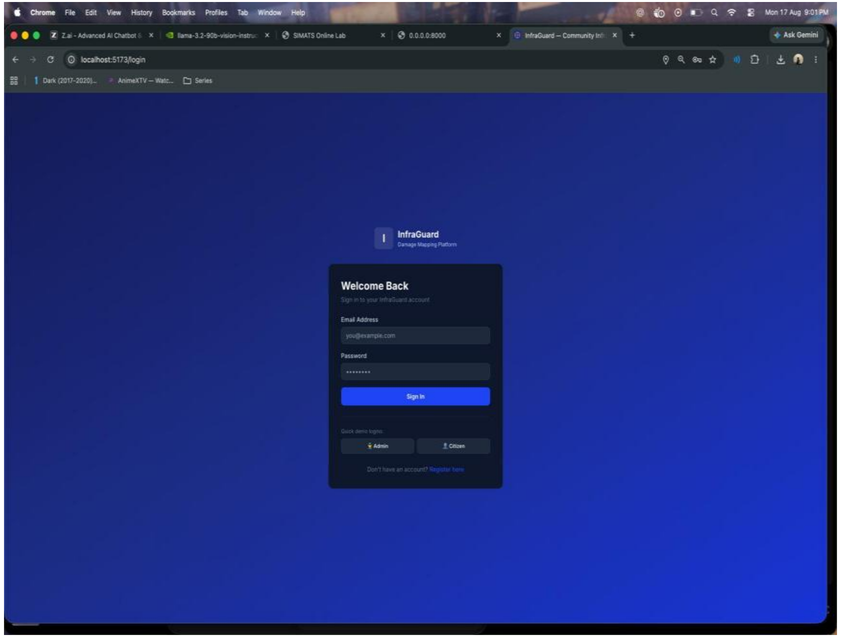
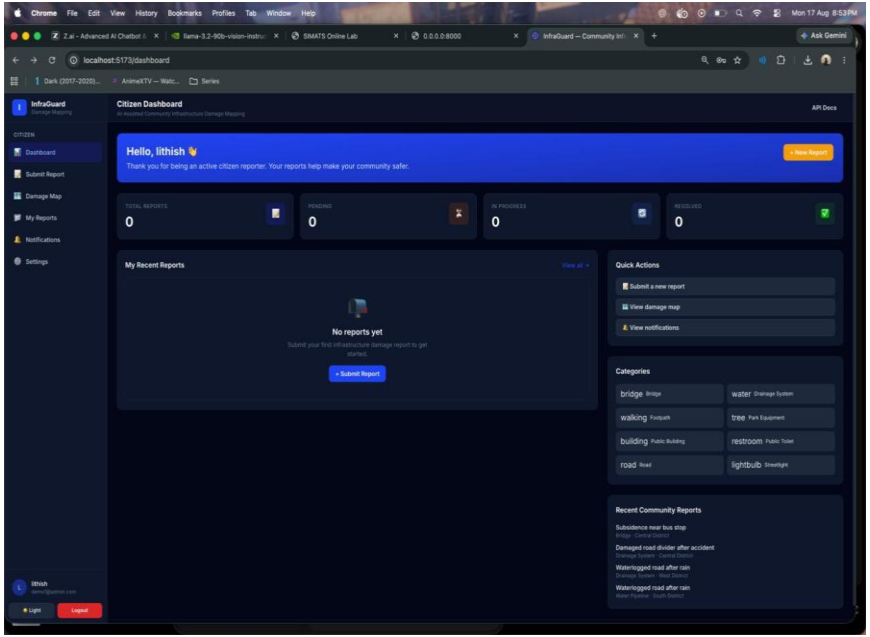
UI


localhost:5173/login
InfraGuard
Damage Mapping Platform
Welcome Back
Sign in to your InfraGuard account
Email Address
you@example.com
Password
Sign In
Admin
Citizen
Don't have an account? Register here


localhost:5173/dashboard
InfraGuard
Citizen Dashboard
API Docs
Dashboard
Submit Report
Damage Map
My Reports
Notifications
Settings
Hello, lithish
Thank you for being an active citizen reporter. Your reports help make your community safer.
+ New Report
TOTAL REPORTS
0
PENDING
0
IN PROGRESS
0
RESOLVED
0
My Recent Reports
No reports yet
+ Submit Report
Quick Actions
Submit a new report
View damage map
View notifications
Categories
bridge
water
walking
tree
building
restroom
road
lightbulb
Recent Community Reports
Subsidence near bus stop
Damaged road divider after accident
Waterlogged road after rain
Waterlogged road after rain
lithish
Logout

Testing

- **Unit Testing**
- JUnit
- Mockito
- User validation
- Report validation
- Severity calculation
- Priority calculation
- **Integration Testing**
- Frontend ↔ Java REST API
- Java ↔ Database
- Report → Severity → Priority
- Authority → Repair Status

- **Test Outputs**
- Login validation
- Issue submission
- GPS/location validation
- Severity calculation
- Priority calculation
- Status update

Challenges

Issue	Solution
GPS accuracy	PostGIS geospatial validation
Image variations	OpenCV preprocessing
Correct severity prediction	Hybrid ML + rule-based approach
Large number of map markers	Marker clustering
Priority calculation	9-factor priority engine

Next Plan

- Complete remaining modules
- Improve UI
- Complete testing
- Fix bugs
- Final deployment & documentation

THANK YOU

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GPS Photo